40-question answer key and explanation guide

REVIEW DRAFT. Source-authored explanations retained; independent academic validation is pending.

01-B 02-A 03-A 04-A 05-A 06-D 07-A 08-B
09-C 10-C 11-B 12-A 13-A 14-B 15-B 16-C
17-D 18-D 19-C 20-B 21-C 22-D 23-C 24-B
25-D 26-D 27-C 28-B 29-A 30-D 31-D 32-D
33-C 34-C 35-C 36-A 37-D 38-A 39-B 40-B

1. An ECG marker is placed at the beginning of the P wave. Where must the second marker go to measure the PR interval?

Answer: B. At the beginning of the QRS complex.

Explanation: The PR interval measures from the onset of atrial depolarisation (P-wave onset) to the onset of ventricular depolarisation (QRS onset).

Not A: The start of T occurs after ventricular depolarisation.

Not C: Including all of QRS would extend beyond the PR interval.

Not D: The end of T is used to mark the end of the QT interval.

Source: lecture-7:Q19 | 6-7

2. Which response becomes particularly important for supporting pressure during sustained rather than momentary upright posture?

Answer: A. Renin-angiotensin effects and renal salt retention.

Explanation: The rapid baroreflex stabilizes pressure within seconds, while the renin-angiotensin-aldosterone system supports longer-term pressure and volume.

Not B: Aldosterone supports sodium retention rather than its suppression.

Not C: Baroreflexes remain available during sustained standing.

Not D: Loss of arteriolar responsiveness would worsen orthostatic control.

Source: circulation-3:Q9 | Circulation 3, slide 4; Chapter 19 ?3

3. A person becomes angry during an argument, and both pulse and pressure rise. Which route can initiate this response?

Answer: A. Cortical signals acting through the hypothalamus on autonomic centres.

Explanation: Emotions such as anger begin in the cortex. The cortex drives the hypothalamus, which excites the vasomotor centre, raising sympathetic output so heart rate and blood pressure rise.

Not B: Renal volume control is too slow and not the primary emotional route.

Not C: The CSF pathways do not directly account for this autonomic response.

Not D: Platelet serotonin is a local vascular constrictor, not the main emotional signal.

Source: circulation-2:Q57 | 30, 33

4. Why can vision fail before consciousness during strong positive acceleration?

Answer: A. Retinal perfusion falls as blood shifts toward the feet.

Explanation: Positive g causes leg pooling and reduced pressure at eye level, so retinal perfusion can fail before cerebral consciousness.

Not B: Pressure at eye level decreases during positive g.

Not C: The mechanism is reduced eye-level arterial perfusion, not an inevitable ICP excess.

Not D: Headward blood movement occurs during negative g.

Source: circulation-3:Q16 | Circulation 3, slide 9; Chapter 19 ?6

5. Which combination correctly characterises isovolumetric ventricular contraction?

Answer: A. All valves closed, ventricular pressure rises, volume fixed.

Explanation: Both ventricles generate rising pressure with all inlet and outlet valves closed, so neither receives nor ejects blood.

Not B: Open valves permit volume exchange.

Not C: AV valves do not remain open once ventricular pressure rises above atrial.

Not D: Falling pressure describes isovolumetric relaxation, not contraction.

Source: circulation-1:Q19 | 14-15

6. A patient loses substantial albumin into urine during kidney disease. What Starling-force change is expected?

Answer: D. Reduced plasma oncotic pressure promotes oedema.

Explanation: Proteinuria reduces circulating albumin and its inward oncotic force, favouring movement of fluid into interstitium.

Not A: Viscosity is not the principal mechanism of protein-loss oedema.

Not B: Capillary permeability elsewhere need not decrease.

Not C: Albumin loss lowers oncotic pressure.

Source: circulation-1:Q88 | 93

7. Why is perfusion of the left ventricular myocardium predominantly diastolic?

Answer: A. Ventricular contraction compresses intramural vessels.

Explanation: High left ventricular wall tension during systole squeezes intramyocardial coronary vessels; relaxation permits better flow.

Not B: Coronary ostia remain open through the cardiac cycle.

Not C: Aortic pressure persists during diastole and drives coronary flow.

Not D: Coronary flow increases during relaxation rather than disappearing.

Source: lecture-6:Q22 | 27; Chapter 23 ?3

8. Beta-1 adrenergic stimulation generally increases ventricular contractility through which sequence?

Answer: B. Increased cAMP/PKA signalling -> greater calcium availability.

Explanation: Beta-1 signalling raises cAMP and PKA activity, enhancing calcium entry and cycling for stronger contraction.

Not A: Blocking release channels would reduce systolic calcium release.

Not C: Vagal slowing is not the main ventricular inotropic pathway.

Not D: Beta-1 stimulation raises, not lowers, cAMP.

Source: circulation-1:Q10 | 46-47; Chapter 2 ?7

9. A person lying still stands up, then begins walking. What feature makes the calf-muscle pump effective?

Answer: C. Deep vein compression with one-way valves.

Explanation: Calf contractions compress deep veins, while competent venous valves favour upward flow toward the heart instead of backflow.

Not A: Venous return requires directed pressure gradients, not dilation alone.

Not B: Capillaries do not function as contractile muscle pumps.

Not D: Arteries do not rely on venous-style one-way valves.

Source: circulation-3:Q6 | Circulation 3, slide 6; Chapter 19 ?3

10. Why may arterial blood pressure rise sharply during severe medullary ischaemia caused by raised intracranial pressure?

Answer: C. Direct vasomotor-centre excitation greatly increases sympathetic outflow.

Explanation: Hypoxic, hypercapnic medullary tissue drives a strong sympathetic pressor response to restore cerebral blood flow.

Not A: Inhibition would tend to lower vascular tone and pressure.

Not B: Vagal discharge slows the heart and is not the primary pressor mechanism.

Not D: ANP favours dilation and volume loss.

Source: circulation-2:Q54 | 31-32, 35-36

11. A standard ECG uses ten skin electrodes but produces twelve electrical views. How is this possible?

Answer: B. Several leads are calculated from combinations of the same electrode voltages.

Explanation: A lead is an electrical viewpoint calculated from one or more electrode potentials, so ten physical contacts can generate twelve different standard recordings.

Not A: Routine twelve-lead ECG does not require intracardiac electrodes.

Not C: All twelve are electrical leads; pressure sensors do not generate extra leads.

Not D: The limb contacts remain in place during a standard recording.

Source: lecture-7:Q14 | 3-5; Chapter 6 ?3

12. Why does positioning an accelerating pilot more horizontally improve tolerance to positive g?

Answer: A. It shortens the heart-to-brain hydrostatic column.

Explanation: A more horizontal posture decreases the vertical distance over which acceleration reduces pressure toward the head and brain.

Not B: Increasing the vertical distance would worsen the gradient.

Not C: Enhanced ankle pooling would reduce central venous return.

Not D: Greater dependent filtration would not protect brain perfusion.

Source: circulation-3:Q17 | Circulation 3, slide 11; Chapter 19 ?6

13. A ventricular depolarisation vector runs almost perpendicular to the viewing axis of a lead. What should that lead record?

Answer: A. A small or biphasic ventricular deflection.

Explanation: Only the component of the electrical vector along the lead axis contributes strongly to the deflection; a near-perpendicular vector has a small projection.

Not B: A strongly downward deflection needs a substantial vector component away from the positive pole.

Not C: A strongly upright deflection needs a substantial vector component towards the positive pole.

Not D: Vector orientation changes amplitude and polarity, not necessarily conduction duration.

Source: lecture-7:Q3 | 3-5; Chapter 6 ?2

14. A vascular bed has a pressure difference of 80 mmHg and resistance of 20 mmHg?min/L. What is its flow?

Answer: B. 4 L/min.

Explanation: Flow equals pressure difference divided by resistance: 80/20 = 4 L/min. Increasing resistance with the same pressure difference would lower flow.

Not A: Two L/min would correspond to resistance of 40.

Not C: Eight L/min would correspond to resistance of 10.

Not D: Three L/min does not fit the given ratio.

Source: circulation-1:Q50 | 61, 67-70

15. In complete AV dissociation, the right atrium occasionally contracts against a closed tricuspid valve. What JVP abnormality may follow?

Answer: B. Intermittent cannon a waves.

Explanation: Unsynchronised atrial contraction against a shut AV valve can produce large intermittent cannon a waves.

Not A: Giant v waves primarily suggest tricuspid regurgitation.

Not C: Atrial contractions still cause pressure variations.

Not D: Venous pulsations can persist despite AV block.

Source: circulation-1:Q72 | 26; Chapter 11 ?7

16. During standing, head-level arterial pressure falls by 30 mmHg and cerebral venous pressure falls by 25 mmHg. How does arterial minus venous driving pressure change?

Answer: C. It falls by about 5 mmHg.

Explanation: The arterial-venous difference changes by -30 minus -25, or -5 mmHg. Falling venous and intracranial pressures therefore help protect cerebral perfusion.

Not A: That ignores the fall in cerebral venous pressure.

Not B: The difference becomes slightly smaller, not larger.

Not D: The changes cannot be added as two rises.

Source: circulation-3:Q7 | Circulation 3, slide 5; Chapter 19 ?3

17. Redness spreads beyond a stroked patch of skin as the flare develops. What generates this spread?

Answer: D. Antidromic impulses in branches of a sensory axon.

Explanation: The flare is an axon reflex: sensory-nerve branches release vasodilator neuropeptides in surrounding skin without a spinal reflex.

Not A: Lymphatic filtration does not produce this arteriolar flare.

Not B: The surrounding redness is functional vasodilation, not thrombosis.

Not C: A central synaptic reflex is not needed for the axon reflex.

Source: lecture-6:Q51 | 40

18. A nurse notices ankle swelling after a long standing shift. Which immediate local change promotes this swelling?

Answer: D. Raised capillary pressure increases outward filtration.

Explanation: The increased hydrostatic pressure in dependent leg capillaries pushes more fluid into the interstitium and promotes oedema.

Not A: A lower capillary pressure would oppose outward filtration.

Not B: Standing promotes dependent filtration; it does not stop capillary exchange.

Not C: Plasma oncotic pressure ordinarily favours fluid retention within blood.

Source: circulation-3:Q4 | Circulation 3, slide 3; Chapter 19 ?2

19. A low-frequency third heart sound occurs in early diastole. What ventricular event coincides with it?

Answer: C. Rapid passive ventricular filling.

Explanation: S3 occurs during rapid early-diastolic ventricular filling, as blood entering the chamber causes vibrations.

Not A: Atrial depolarisation precedes the atrial contribution later in diastole.

Not B: Isovolumetric contraction occurs near S1.

Not D: Aortic valve opening occurs during ventricular systole.

Source: circulation-1:Q29 | 28

20. A small additional deflection appears after the T wave in an otherwise interpretable trace. Which description fits the U wave?

Answer: B. A possible late ventricular repolarisation phenomenon.

Explanation: The lecture describes a U wave, when seen after T, as the final remnants of ventricular repolarisation. Its precise physiological origin is uncertain.

Not A: Atrial depolarisation is the P wave preceding QRS.

Not C: Rapid ventricular depolarisation is represented by QRS, not U.

Not D: The ventricular plateau is associated with the ST segment.

Source: lecture-7:Q29 | 10

21. What is the main purpose of wide changes in skin blood flow during heating or cooling?

Answer: C. To adjust body heat gain and loss.

Explanation: Cutaneous vessels dilate to increase heat loss and constrict to conserve heat; flow can far exceed nutritional needs.

Not A: Skin metabolic demand is relatively modest compared with variation in perfusion.

Not B: Heart pacemaking is not the main thermoregulatory role of skin circulation.

Not D: Pulmonary-systemic pressure equality is unrelated to thermoregulation.

Source: lecture-6:Q46 | 39-41

22. Why does sustained systemic hypertension tend to raise myocardial oxygen demand?

Answer: D. The ventricle must generate higher pressure for ejection.

Explanation: Higher arterial pressure increases ventricular pressure work and wall stress, requiring more oxygen for force generation.

Not A: Higher afterload generally increases wall stress.

Not B: The myocardium continues to contract against the higher load.

Not C: Coronary oxygen extraction remains substantial.

Source: circulation-1:Q48 | 54; Chapter 8 ?7

23. Why is a distinct atrial repolarisation wave usually absent from a normal ECG?

Answer: C. Its small electrical signal is obscured by the ventricular QRS complex.

Explanation: Atrial repolarisation occurs around the time of ventricular depolarisation; the much larger QRS electrical signal masks it on the surface trace.

Not A: Repolarisation does generate voltage differences, although the signal is small.

Not B: Atrial repolarisation does not need to await the beginning of ventricular diastole.

Not D: That statement does not explain why the wave is not separately visible.

Source: lecture-7:Q25 | Chapter 6 ?4 (approved-chapter explanation)

24. A measured QRS occupies 3.5 small squares at 25 mm/s. Which statement is supported by the lecture's usual duration range?

Answer: B. It lasts 0.14 s and is longer than the stated usual range.

Explanation: At 25 mm/s, 3.5 small squares equal 3.5 x 0.04 = 0.14 s, above the lecture's usual QRS duration of 0.06-0.10 s.

Not A: 0.10 s corresponds to 2.5 small squares at the standard speed.

Not C: 0.07 s incorrectly halves the 0.14-s measured duration.

Not D: 0.35 s does not follow from the standard 0.04-s small-square duration.

Source: lecture-7:Q39 | 8, 13

25. A patient becomes dizzy on standing after starting a sympatholytic drug. Which compensatory response is most directly weakened?

Answer: D. Constriction of systemic arterioles and veins.

Explanation: Sympatholytic drugs blunt reflex arteriolar and venous constriction, so the circulation cannot adequately oppose blood pooling and falling arterial pressure on standing.

Not A: Walking uses a mechanical muscle pump, not this autonomic drug pathway.

Not B: Venous valve directionality is mechanical and not directly abolished by sympatholytic drugs.

Not C: Cerebral metabolic vasodilatation is not directly a sympathetic vasoconstrictor response.

Source: circulation-3:Q15 | Circulation 3, slide 8; Chapter 19 ?5

26. A patient develops rapidly increasing intracranial pressure after a brain haemorrhage. What first threatens cerebral perfusion?

Answer: D. Compression of brain vessels and reduced driving pressure.

Explanation: Rising intracranial pressure compresses intracranial vessels and lowers cerebral perfusion pressure, risking medullary ischaemia.

Not A: Raised intracranial pressure does not require systemic blood loss.

Not B: Dilation cannot prevent the loss of cerebral driving pressure as intracranial pressure rises.

Not C: Perfusion initially becomes harder as intracranial pressure rises.

Source: circulation-2:Q53 | 31-32, 35-36

27. What closes the aortic valve at the beginning of ventricular relaxation?

Answer: C. Aortic pressure exceeds ventricular pressure.

Explanation: When ventricular pressure falls below aortic pressure, brief reversal tendency closes the aortic valve and starts isovolumetric relaxation.

Not A: Venous pressure does not directly determine aortic valve closure.

Not B: Atrial pressure governs mitral opening, not aortic closure.

Not D: Ventricular pressure exceeding aortic pressure opens the valve.

Source: circulation-1:Q23 | 17

28. Which sequence best describes the classical triple response of Lewis?

Answer: B. Red line, flare and wheal.

Explanation: The classical triple response consists of local redness, a spreading flare and an oedematous wheal.

Not A: A white line is a separate lighter-stroke reaction.

Not C: Wheal formation requires local fluid leak rather than none.

Not D: Thrombosis and scar are not normal features.

Source: lecture-6:Q52 | 40

29. A patient has had high arterial pressure for months. Why might baroreceptors fail to drive pressure back to the original baseline?

Answer: A. Their operating range resets around the sustained level.

Explanation: When pressure stays high for days, the baroreceptors reset and treat the new level as normal, so they defend it instead of correcting it. Long-term pressure is set mainly by the kidneys' control of salt and water.

Not B: Unbounded sensitivity would strongly oppose the persistent rise.

Not C: Sensory fibres do not become motor fibres.

Not D: Resetting does not require loss of every afferent signal.

Source: circulation-2:Q44 | 18, 24-27

30. A very light stroke briefly leaves a pale line on the skin. How is this reaction best classified?

Answer: D. A brief local vasoconstrictor response.

Explanation: The lecture describes a transient white reaction due to constriction; it is distinct from the classical red-flare-wheal response.

Not A: Wheal is raised local oedema from permeability.
Not B: The red line is a separate vasodilator response.
Not C: Flare is surrounding redness from arteriolar dilation.

Source: lecture-6:Q49 | 39-40

31. How can fluid enter a blind-ended lymphatic capillary when interstitial pressure rises?

Answer: D. Overlapping endothelial flaps admit interstitial fluid.

Explanation: Initial lymphatic endothelial flaps allow interstitial fluid and proteins to enter when surrounding pressure exceeds lymphatic pressure.

Not A: Initial lymphatic capillaries lack a thick continuous smooth-muscle coat.
Not B: There is no single central lymphatic heart.
Not C: Initial lymphatics are relatively permeable to interstitial proteins.

Source: circulation-1:Q93 | 60; Chapter 15 ?2

32. If the sinoatrial node fails, why may the heart still beat at a slower rate?

Answer: D. Other conducting tissues retain latent automaticity.

Explanation: Subsidiary pacemakers in the AV junction or Purkinje system can spontaneously depolarise, usually at slower rates.

Not A: Autonomic nerves modify pacemaking but do not directly activate every myocyte.
Not B: Coordinated contraction still requires electrical activation.
Not C: Valves are passive mechanical structures.

Source: circulation-1:Q13 | 7; Chapter 3 ?4

33. A patient with raised intracranial pressure develops hypertension and bradycardia together. What best explains the slow pulse?

Answer: C. The arterial baroreflex reacts to the sympathetic pressure rise.

Explanation: Medullary ischaemia first raises sympathetic pressure; arterial baroreceptors then increase vagal drive and slow the heart.

Not A: beta1 stimulation would usually accelerate, not slow, the heart.
Not B: Cardiac autonomic signalling remains central to the response.
Not D: Atrial volume receptors respond chiefly to filling, not this sequence.

Source: circulation-2:Q55 | 32-36

34. Blood flow to the medullary vasomotor centre becomes severely reduced. Which local chemical change directly activates its sympathetic response?

Answer: C. Hypoxia together with accumulating carbon dioxide.

Explanation: Poor medullary perfusion produces hypoxia and hypercapnia, which strongly stimulate the vasomotor centre to raise pressure.

Not A: High oxygen and low CO2 do not represent medullary ischaemia.
Not B: Low acidity and abundant oxygen oppose the described stimulus.
Not D: Adequate oxygen delivery would not produce the ischaemic response.

Source: circulation-2:Q52 | 30-32

35. Blood ejected from the right ventricle returns to the left atrium through which sequence?

Answer: C. Pulmonary artery -> lung capillaries -> pulmonary veins.

Explanation: The right ventricle pumps blood through pulmonary arteries to lung capillaries; pulmonary veins return it to the left atrium.

Not A: The aorta receives systemic output from the left ventricle.

Not B: That is mainly the systemic circulation after left ventricular ejection.

Not D: Pulmonary veins carry blood into the left atrium rather than away from the right ventricle.

Source: circulation-1:Q1 | 2-6

36. Arterial pressure increases suddenly and baroreceptor firing rises. What changes first in central reflex control?

Answer: A. NTS activation increases depressor and vagal signalling.

Explanation: More stretch -> faster baroreceptor firing -> more input to the NTS. The NTS excites the depressor area (which inhibits the pressor area) and the vagal neurons, so sympathetic output falls and vagal output rises.

Not B: The heart remains under autonomic regulation during the reflex.

Not C: The reflex suppresses rather than increases sympathetic pressor drive.

Not D: Increased wall stretch increases, rather than decreases, afferent traffic.

Source: circulation-2:Q40 | 22-24

37. A sudden increase in sympathetic venous tone can support cardiac output mainly through which immediate effect?

Answer: D. Venoconstriction shifts stored blood centrally.

Explanation: Sympathetic venoconstriction reduces the capacity of the venous reservoir and shifts stored blood toward the heart, supporting venous return and end-diastolic filling.

Not A: Systemic arterioles have no valves that close with sympathetic stimulation.

Not B: Venodilation generally increases peripheral capacitance and pooling.

Not C: Venoconstriction redistributes existing blood rather than rapidly generating plasma.

Source: circulation-1:Q73 | 73-75, 97-98; Chapter 12 ?2

38. What most clearly differentiates myocardial infarction from brief stable angina?

Answer: A. Prolonged ischaemia can produce myocyte necrosis.

Explanation: Prolonged severe ischaemia can cause irreversible myocardial cell death; transient angina does not necessarily cause necrosis.

Not B: Infarction can occur across a range of heart rates.

Not C: Transient angina ordinarily reflects reversible myocardial ischaemia.

Not D: Stable angina may result from fixed stenosis without rupture.

Source: lecture-6:Q28 | 29

39. What helps maintain forward systemic arterial flow between heartbeats?

Answer: B. The elastic recoil of large arteries sustains diastolic flow.

Explanation: Elastic arteries stretch during systole and recoil during diastole, smoothing pulsatile ejection into continuous downstream flow.

Not A: Venous valves prevent backflow but do not create aortic pressure.

Not C: Respiratory movement mainly assists venous return.

Not D: Capillaries lack the pumping muscle required to drive systemic flow.

Source: circulation-1:Q49 | 55-57

40. A patient with coronary atheroma is comfortable at rest but develops exertional chest pain. Which explanation fits best?

Answer: B. Restricted coronary flow reserve cannot meet increased demand.

Explanation: A narrowed vessel may deliver adequate resting flow but cannot dilate enough to meet exercise demand, causing reversible ischaemia.

Not A: Plaque does not disappear with exertion.

Not C: Exercise raises myocardial work and oxygen requirements.

Not D: Resting coronary flow may be sufficient despite reduced reserve.

Source: lecture-6:Q27 | 29; Chapter 23 ?6